

Project Title: Offline RL for Adaptive Vocabulary Selection in Conversational Language Tutoring**Team Members:** Onyinyechi Okoye**Emails:** onyie@stanford.edu**1 Objective**

Conversational language tutoring applications powered by large language models face a fundamental decision at every turn: which vocabulary should the tutor’s response target? Each turn presents a mixture of words the learner has never seen, words currently being acquired, and words due for spaced-repetition review. The tutor must balance introducing new material against reinforcing old, all while producing natural-sounding dialogue. This is a sequential decision problem with delayed, multi-objective rewards: the effect of introducing a word now on long-term retention is not observed until future sessions, and optimizing purely for retention can produce stilted conversation that undermines engagement.

We propose formulating vocabulary selection as a Markov decision process and training an offline RL policy that outputs structured vocabulary directives to a conversation-generating LLM, treating the LLM not as the policy but as a stochastic component of the environment through which the policy’s decisions are realized. Our objective is to learn a policy that outperforms both traditional spaced repetition schedulers like FSRS [1] and direct LLM-prompting on cumulative vocabulary retention while maintaining conversational naturalness. We hypothesize that the learned policy will discover nontrivial scheduling strategies that differ from fixed heuristics, such as adaptive pacing of new vocabulary based on learner state or context-dependent mixing of review and introduction that a hand-tuned rule cannot anticipate. We scope the project to a fixed scenario, a single AI character, and a vocabulary of approximately 500 words, with a knowledge-tracing-based student simulator standing in for real learners.

2 Related Work

Spaced Repetition Scheduling. The SM-2 algorithm [2] adjusts review intervals based on per-item difficulty and recall performance, and remains the foundation of most flashcard systems. The Free Spaced Repetition Scheduler (FSRS) [1] improves on SM-2 by formulating scheduling as a stochastic shortest path problem, achieving strong retention across millions of Anki users. However, both assume an isolated-item review paradigm where the scheduler shows a flashcard, the user rates recall, and the interval updates. They have no mechanism for embedding review into natural conversation or jointly handling the introduction of new vocabulary alongside review.

Knowledge Tracing. Deep Knowledge Tracing [3] models a student’s evolving knowledge state as a hidden state in a recurrent network, predicting future recall probabilities given interaction history. We draw on this literature to build a student simulator that models per-word skill evolution with exposure, serving as the training and evaluation environment for RL policies.

RL for Adaptive Teaching. Doroudi et al. [4] survey RL for educational problems and identify reward specification as a central challenge. Rafferty et al. [5] formulate teaching as POMDP planning, showing model-based policies can accelerate concept learning. However, this work operates in discrete, structured environments (e.g., selecting which problem to assign). None addresses the setting where item selection must be realized through a generative language model producing free-form text, which introduces a naturalness constraint absent from traditional educational RL.

RL with Language Models. RLHF [6] trains LLMs to follow instructions by optimizing a learned reward via PPO [7], treating the LLM as the policy. Our setting differs: the LLM is a component of the *environment*, not the policy. The RL policy selects vocabulary directives, and a frozen conversation LLM executes them. This creates a meta-level control problem where the policy must learn what directives produce good outcomes when mediated through a stochastic generative model. More precisely, we introduce a hierarchical control setting in which RL operates over latent linguistic constraints that are executed by a generative model, combining the scheduling strengths of spaced repetition with the contextual flexibility of LLM generation in a way that neither approach achieves alone.

3 Technical Outline

MDP Formulation. The state consists of: (1) the learner’s vocabulary profile (per-word FSRS stability, difficulty, retrievability, and usage counts, compressed into a fixed-dimensional vector); (2) the scenario context (topic, register, conversation length); and (3) in-session exposure history. The action is a structured vocabulary directive specifying the number of new, actively-learning, and review words to target, plus a specific emphasized word. To keep the action space tractable, we start with this simplified structure rather than a full combinatorial ranking, though we plan to explore richer action spaces (e.g., top- k word selection or distributions over candidates) if the simplified version proves too constrained. The conversation LLM receives the directive in its prompt and generates a character response accordingly. The reward combines: (1) a naturalness score from an LLM-as-judge evaluator that rates whether a turn sounds like plausible human dialogue given the scenario, with lightweight proxies (perplexity, repetition penalties) used in compute-limited ablations; (2) short-term retention when the word reappears later in session; and (3) longer-term retention measured in subsequent simulated sessions.

Student Simulator. We build a knowledge-tracing simulator over approximately 500 Spanish words spanning multiple CEFR levels. Each word has a latent skill state evolving according to an FSRS-inspired forgetting curve: exposure increases stability, elapsed time decreases retrievability, and difficulty modulates the learning rate. The simulator takes a directive and the LLM-generated turn as input, probabilistically simulates the learner’s comprehension based on current retrievability, and updates skill states. Because the central risk of simulator-based evaluation is that the learned policy overfits simulator dynamics, we take several steps to guard against this. We introduce stochastic noise into learner behavior (variable attention, inconsistent recall), train on one family of simulator parameterizations and evaluate on held-out configurations with different forgetting rates and difficulty distributions, and intentionally misspecify the simulator relative to the model the policy was trained on. If the learned policy degrades gracefully under misspecification rather than collapsing, it provides evidence that the policy has learned generalizable structure rather than simulator-specific shortcuts.

Policy and Training. The policy is a feature-based MLP over precomputed vocabulary statistics. We collect trajectories from baseline policies interacting with the simulator, then train using Implicit Q-Learning (IQL) [8], which avoids querying out-of-distribution actions and handles mixed-quality offline data. We compare against Conservative Q-Learning (CQL) [9] as a secondary method.

Baselines and Evaluation. We compare against: (1) pure FSRS selecting top- N due items; (2) FSRS plus one new word per turn; (3) a rule-based 95%/5% heuristic; (4) LLM-prompted selection given the full vocabulary state; and (5) random. All baselines use the same conversation LLM. We evaluate on cumulative vocabulary gain over multi-session trajectories, per-word retention curves, naturalness scores via LLM judge, and an engagement proxy. We include ablations removing the naturalness reward, the multi-category action structure, and varying simulator parameters. Beyond aggregate metrics, we will visualize sample conversations and per-word vocabulary trajectories under each policy to qualitatively characterize the scheduling strategies the learned policy discovers.

4 Project Phases

Phase 1: Simulator and Infrastructure. Build the knowledge-tracing student simulator, define the vocabulary inventory and FSRS skill dynamics, implement the conversation LLM interface with directive-following prompt templates, and collect baseline trajectories from FSRS, rule-based, LLM-prompted, and random policies.

Phase 2: Policy Training. Implement the MDP formulation and state featurization, train IQL and CQL policies on offline trajectory data, iterate on reward component weights and action space design, and run initial comparisons against baselines.

Phase 3: Evaluation and Writing. Run full evaluation across all metrics (retention, naturalness, engagement), perform ablation studies and simulator sensitivity analyses, and write the final report.

References

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